

Proof

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Definition of Big-O Notation

Definition (Big-O). A function $f(n)$ is $O(g(n))$ if there exist constants $M > 0$ and n_0 such that

$$f(n) \leq M g(n) \quad \text{for all } n \geq n_0.$$

Why Binary Search is $O(\log n)$

- ▶ Each iteration of binary search cuts the search interval in **half**.
- ▶ After k iterations, the remaining interval size is at most:

$$\frac{n}{2^k}.$$

- ▶ The algorithm stops once the interval size < 1 :

$$\frac{n}{2^k} < 1 \Rightarrow 2^k > n \Rightarrow k > \log_2 n.$$

- ▶ Therefore the worst-case number of iterations satisfies:

$$T(n) \leq \lfloor \log_2 n \rfloor + 1 = \frac{\log n}{\log 2} + 1 = O(\log n).$$

1

¹I am not entirely sure whether you will encounter this proof in the exam (personally, I do not thank so, as we have many other things that are more interesting to test).